Reference-gauge Figure-3 same-tree resolution investigation

AthenaK Z4c / native vertex-centered Cartoon SO(2)

29 August 2026

Verdict: Exact replay is verified and the fine triple is O4-compatible in central curvature and lapse through the first peak. N1024 strongly reduces the peak C/H/M constraints relative to N512, but constraint convergence is nonuniform and Z barely improves. The combined CFL 0.40 / KO 0.10 N1024 ablation fails before the first peak and is not a remedy.

1 Controlled experiment

N128, N256, N512, and baseline N1024 use the same root MeshBlock layout, physical MeshBlock bounds, accepted LogicalLocation hierarchy, and AMR event times. Only cells per physical MeshBlock change in the sequence 16^2, 32^2, 64^2, and 128^2. All use shock-avoiding Bona–Masso lapse with $\kappa = 1$, unit initial lapse, prescribed zero shift, native vertex-centered Cartoon SO(2), O4 finite differences, q6 prolongation, RK4, CFL 0.15, KO dissipation 0.50, zero Z4c constraint damping, and outer boundary $128M$.

The N1024 ablation uses the same N1024 state, hierarchy, and numerical method, but changes *both* CFL to 0.40 and KO dissipation to 0.10. It is therefore a test of that combined setting, not a one-parameter sensitivity measurement.

The hierarchy authority SHA-256 is `7055de601e6181e5ad7e1432b5c20a111b0ba67e0e8d5377c170ea80e7bedcde`. Every replay accepted the two events at their exact physical times and ended with 212 leaves at physical level 5.

2 Execution and provenance

N128–N512 ran on Aurora PVC. Baseline N1024 and the ablation ran on four Perlmutter A100 GPUs using source `02704c79ffe95312cfeba9acde3d38f8b9677dec`, source tree `9605962eed2b74fba6f1d70e6fc526a8f5f56bba`, and executable SHA-256 `a13a58be5e45b37c4c29f65eab2f6fa0d43f8d235d0c51e7ed3459b18e9e247e`. A matched N512 interval agrees between Aurora and Perlmutter at backend roundoff scale.

Case	final coordinate time	final proper time	disposition
N128	45.0000	19.33240	reached tlim; constraint invalid
N256	30.0000	11.28631	reached tlim; constraint invalid
N512	38.6523	14.98253	endpoint; constraint invalid at peak
N1024 baseline	38.6523	14.98066	reached tlim; constraint invalid at peak
N1024 CFL 0.40 / KO 0.10	16.7247	8.68734	failed state gate

Baseline N1024 was Perlmutter job 57702588 and completed in 02:37:46 with clean Athena, launcher, and scheduler exits. The ablation was job 57706639. An earlier job, 57706621, failed before science initialization because the runner used the invalid key `time/cfl`; it produced no science data.

3 Direct Figure-3 comparison

No time, amplitude, or vertical transform was fitted.

Case	peak τ	peak $\log_{10}\lvert\mathcal{K}\rvert$	peak $\int C^2\, dV$
N128	10.31396	4.29765	107.608
N256	10.30333	5.01349	48.2330
N512	10.30811	5.38112	4.09930
N1024 baseline	10.30964	5.47867	0.0388727
Published	10.30683–10.31384	5.47778–5.48688	–

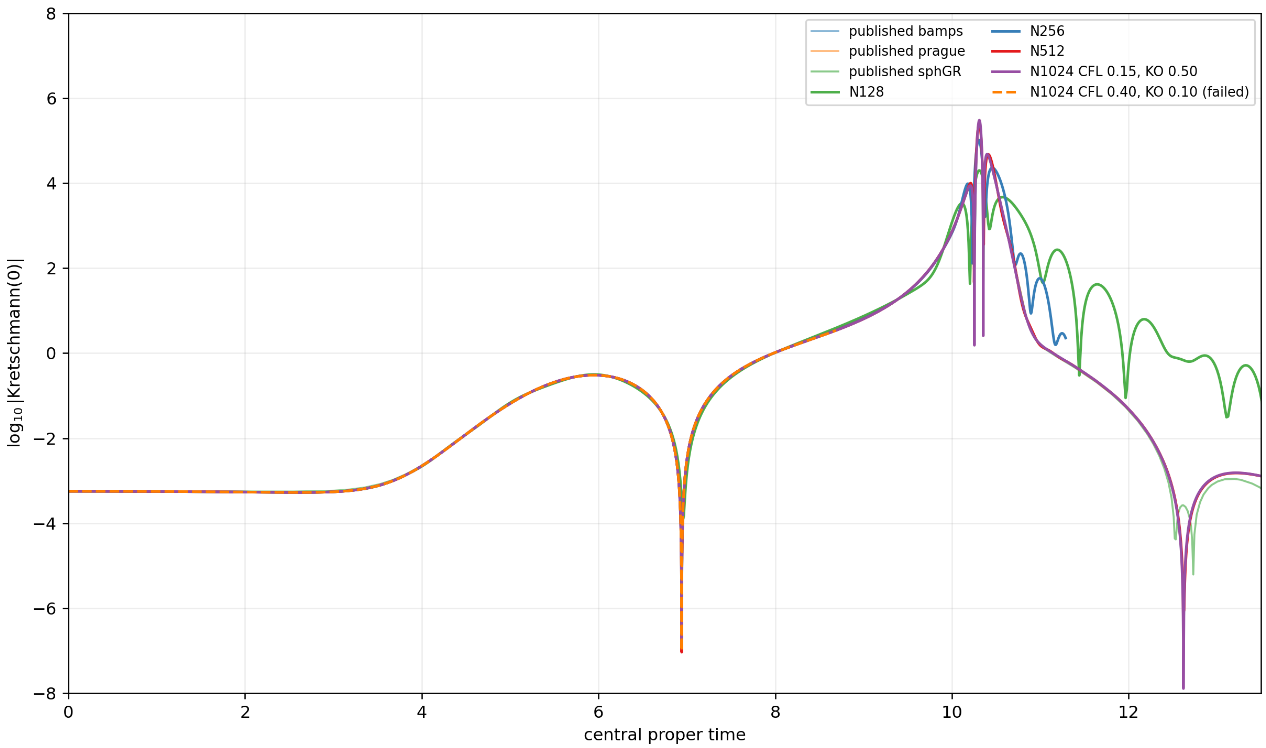

Figure 1: Central Kretschmann comparison. The orange dashed ablation is a failed partial trajectory and ends at $\tau = 8.687$, before the first peak.

Baseline N1024 directly matches the published first peak. Its deep minimum is at $(\tau, \log_{10}\lvert\mathcal{K}\rvert) = (12.61950, -7.88841)$, deeper than the published band, while the rebound at $(13.21271, -2.81258)$ lies in the band. There is no three-resolution late-time convergence claim because N256 ends at $\tau = 11.286$.

4 Constraint evidence

The history measure is already the physical axisymmetric ring measure,

$$2\pi\rho\sqrt{|\det\gamma|}\,d\rho\,dz,$$

with canonical vertex ownership and trapezoid endpoint weights. The reported behavior is not a fictitious collapsed-y normalization artifact.

Squared integral	N128 max	N256 max	N512 max	N1024 max
C	107.608	48.2330	4.09930	0.0388727
H	88.4516	41.1314	3.63079	0.0315101
M	25.3663	9.53227	0.682824	0.00453466
Z	0.643210	0.0550105	0.00462028	0.00400120

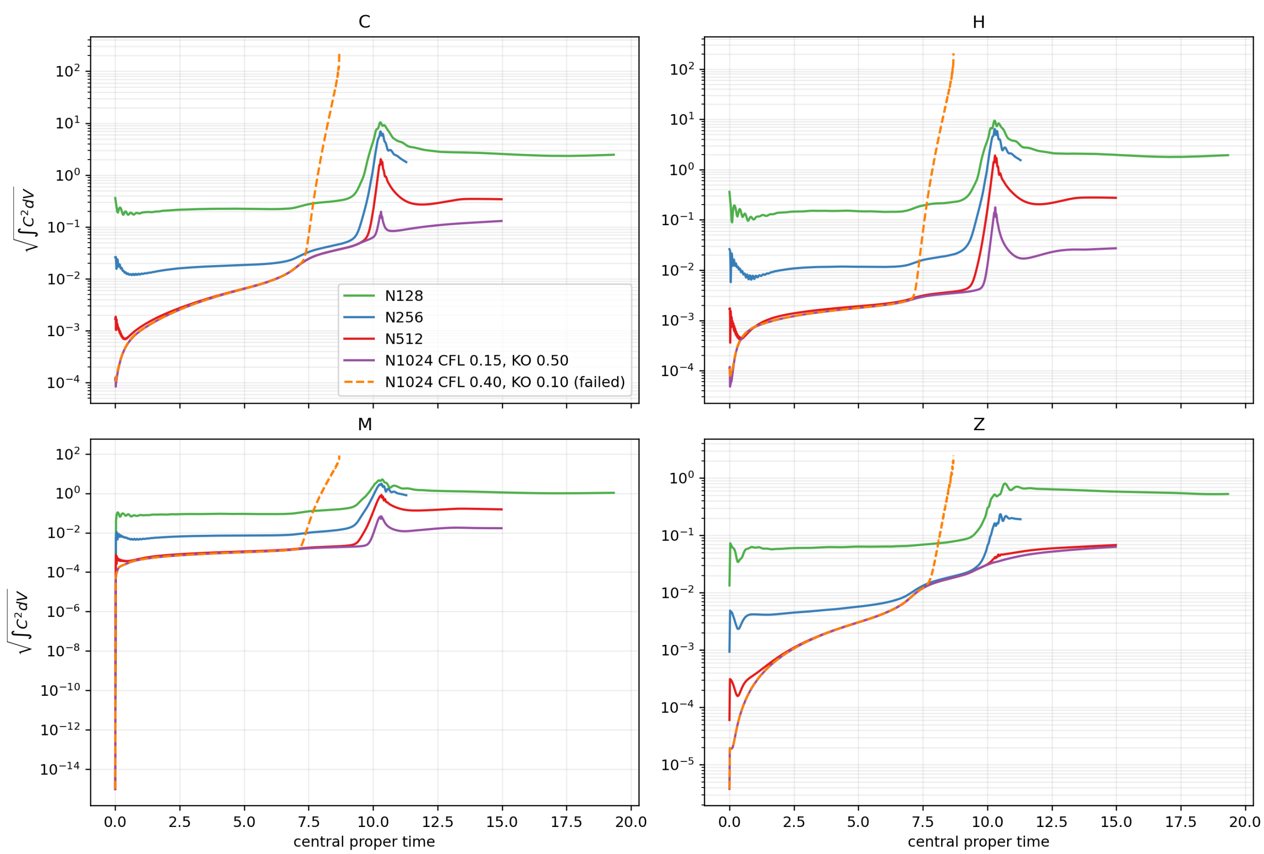

Figure 2: Physical-volume constraint amplitudes. The failed CFL/KO ablation runs away by orders of magnitude after $\tau \simeq 7.5$.

The initial C squared integrals for N128, N256, N512, and N1024 are 1.30505×10^{-1}, 6.85048×10^{-4}, 2.88883×10^{-6}, and 1.28813×10^{-8}. The N512/N1024 initial-amplitude order is 3.90. Initial data resolution is therefore not the source of the later loss of uniform constraint convergence.

The N512/N1024 median amplitude orders over $0<\tau<8$ are 0.007 (C), 0.081 (H), 0.111 (M), and 0.001 (Z): after the initially convergent slice, both fine runs sit on a shared evolution/diagnostic floor. Over $10<\tau<11.286$, the orders rise to 2.69, 4.00, 3.74, and 0.27. C/H/M therefore improve during collapse, while Z shows little useful fine-pair convergence. There is no single constraint order valid across families and times.

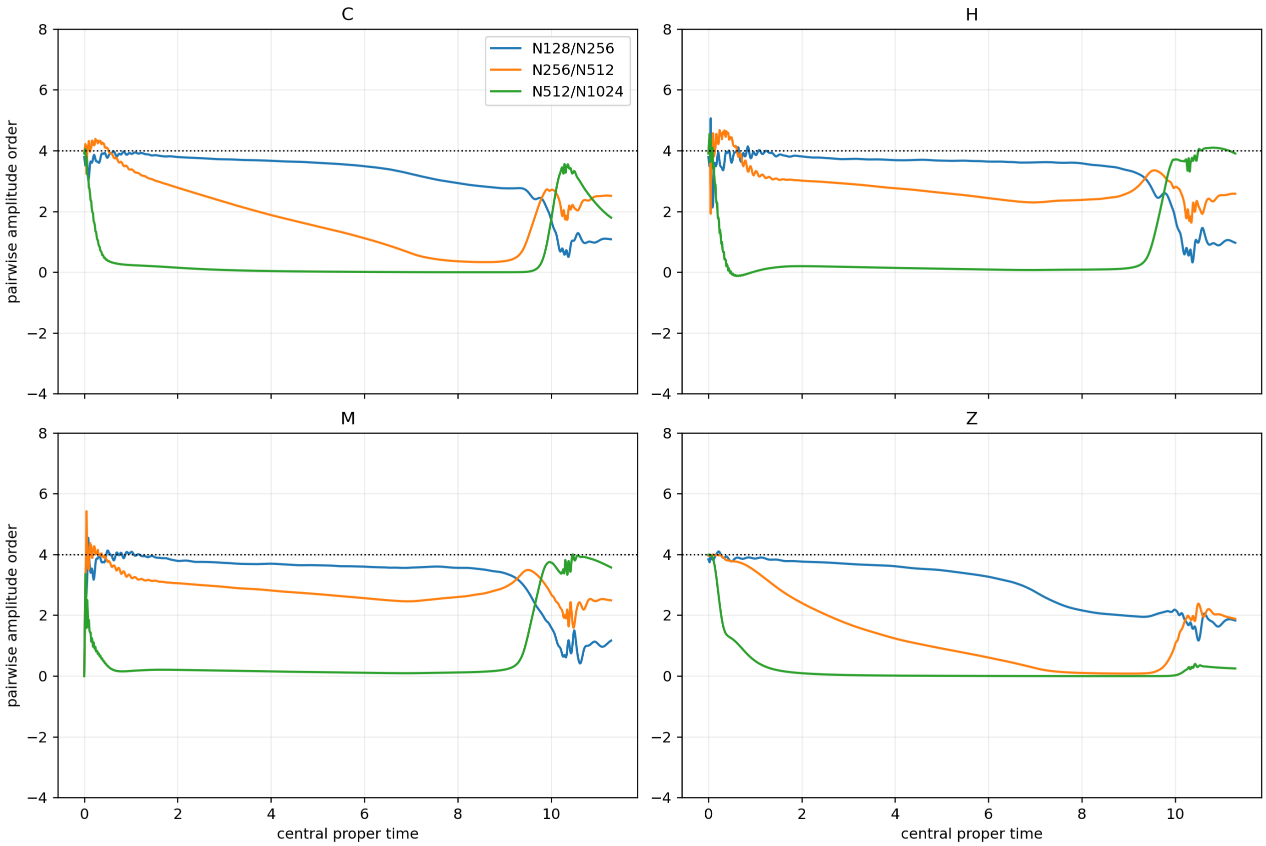

Figure 3: Pairwise amplitude orders for the four-resolution sequence. The ablation is excluded because it is the same resolution with changed CFL and KO, not a fifth resolution.

5 CFL 0.40 / KO 0.10 failure

Baseline and ablation initial constraints are bitwise-identical in the history output. The ablation C squared integral crosses 0.01, 1, 100, and 10000 at $\tau = 7.54619, 7.85416, 8.24578$, and 8.65239, respectively. The last history row has $\int C^2 dV = 4.9679 \times 10^4$, corresponding to amplitude 222.89, versus a baseline maximum of 0.197.

The strict state-admissibility diagnostic stops the run at coordinate time $16.7246503M$, cycle 4919, RK stage 1. The first invalid active point is near the axis at $(\rho, z) = (0.0625, 19.96875)$. Chi is still positive at 0.91660, but the conformal metric has

$$\det \tilde{\gamma} = -0.29184, \qquad (p_0, p_1, p_2) = (18.8750, -0.23072, -0.29184),$$

so it is no longer positive definite. This is an evolved-state failure after a constraint runaway, not a nonpositive-chi prolongation-parent failure.

The result rules out the *combined* CFL 0.40 / KO 0.10 choice as a cure. It cannot say whether the larger time step, weaker dissipation, or their interaction is responsible. Any causal follow-up must vary only one control at a time.

6 Interpretation

The baseline results do not show a simple loss of convergence from N512 to N1024. Peak C/H/M amplitudes fall by factors of about 10–12 and the central first peak approaches the published curve at O4-compatible order. What fails is *uniform* constraint convergence: a shared early floor develops, Z barely improves, and the strict C gate is still exceeded near collapse.

This combination points away from insufficient initial-data resolution and toward an error generated during evolution. Candidate mechanisms include an axis/regularity truncation floor, a persistent AMR/interface contribution, or a formulation/gauge mode whose coefficient does not scale like the bulk O4 error. Existing evidence does not isolate one of them.

The smallest useful next step is a bounded N1024 localization audit around the onset and peak of the residual constraint floor, retaining the exact tree and baseline CFL/KO. It should separate active-axis stencils, same-level seams, and coarse-fine/interface regions. Broad tuning would be less diagnostic.

7 Claim boundary

Supported: exact four-resolution replay; high-order-convergent initial constraints; fine-triple O4-compatible central fields through the first peak; large N512-to-N1024 peak C/H/M improvement; direct N1024 first-peak agreement; and early catastrophic failure of the combined CFL 0.40 / KO 0.10 setting.

Not supported: a constraint-qualified Figure-3 reproduction; a uniform constraint order; three-level convergence across the late minimum and rebound; attribution of the ablation failure to CFL or KO individually; exclusion of axis/interface mechanisms; or a unique source-level defect.